Fig. 1: Ghazipur landfill in New Delhi (Credit: www.firstpost.com)

tragedy. What did we learn from this? Let us delve into it and uncover some facts related to e-waste.

Facts and figures of e-waste

Plastic is present in all electronic products. It is also found in most other things and places. This is where the problem comes in, in the form of pollution. It takes more than 500 years for plastic to decompose. e-Waste, or electronic/electrical waste, includes discarded computers, pen drives, cellphones, televisions and refrigerators. Discarded electronics contain potentially-harmful materials such as lead, cadmium, beryllium or brominated flame retardants.

Generally, e-waste is categorised into five groups based on materials used: ferrous metals, non-ferrous metals, glass, plastic and others. Nearly 48 per cent of the total e-waste is iron and steel, 21 per cent is plastics, 13 per cent is non-ferrous metals including precious metals of which nearly seven per cent is copper.

Problems of trash and e-waste are not only prevalent in India; these are everywhere including the US. The US and China are the two biggest producers of e-waste in the world. Whereas, India is the fifth largest e-waste producer in the world.

Modern industrialisation, population growth and rapidly changing urban lifestyles lead to a myriad of pollution problems including industrial e-waste and garbage that are threatening the lives of all creatures on the planet. In the context of the current industrial revolution, Big Data, big e-waste and big trash are almost synonymous.

e-Waste materials are illegally dumped worldwide. Some reports say that 90 per cent of electronic devices are simply thrown away and mostly end up in landfills, and only 10 per cent of these devices are recycled. Report also says that 70 per cent of the toxicants present in most landfills comes from e-waste. But then, why do I care?

It is a fact that in the region where there is no waste management, apart from various pollutions, diseases are likely to increase. This is because bugs and rats are likely to move in. Improper waste management impacts health and environment through soil, land, water and air. Toxic chemicals from landfills can spread to nearby areas, flow into rivers, emit greenhouse gases and contaminate the water table underground. Landfills create a toxic legacy for future generations. A representation of a landfill is shown in Fig. 2.

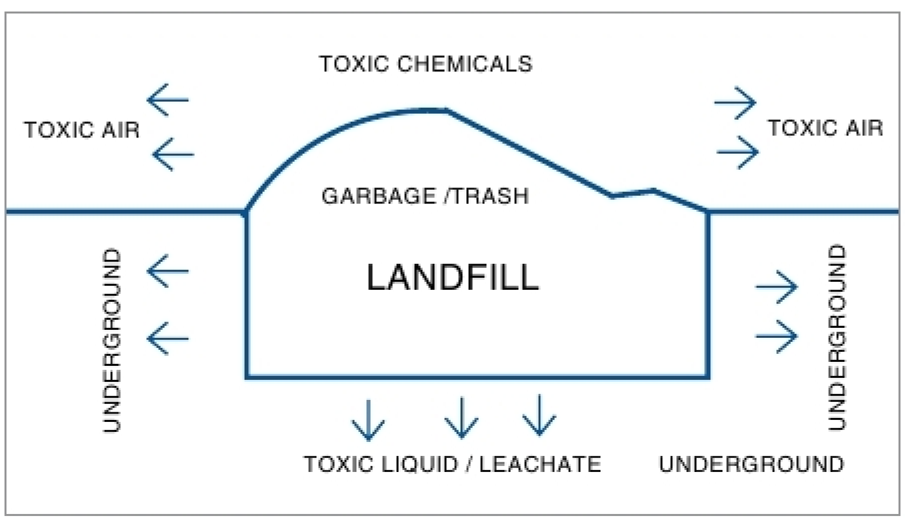

Fig. 2: Representation of toxic chemicals from a landfill

Scope of e-waste recycling and management

e-Waste recycling is a multibillion-dollar business. Gold as well as other precious metals make up a significant amount of e-waste. Hence, extracting these precious materials from e-waste has proven to be advantageous, especially for the gold industry.

Some people separate different electronic components from e-waste and recycle these for resale. Remaining parts of e-waste, which cannot be recycled or are worthless, are either sent to landfills or burnt (incinerated). Some countries import waste materials to generate electricity. This is because burning waste materials under proper conditions can help produce electricity.

As the old saying goes, "One man's trash is another man's treasure." From the moment you throw waste into the dustbin and it lands up in a landfill, there are huge business benefits involved. Waste management is a lucrative business, and there is a lot of money to be made from waste materials. Waste recycling businesses are managed by waste management giants, including Amazon.

The goal of Zero Waste movement that started around a decade ago is to encourage people to reuse products, so that no trash is sent to landfills, incinerators or the ocean. That is, to save our health and environment, we should try to reduce e-waste, ideally to zero. Sweden has almost zero landfills.

We should try to reduce e-waste, reuse and recycle. As a last resort, only materials that cannot be recycled should be either burned (incinerated) or sent to landfills.

Buying a gadget is easy; you can get it from any shop, but recycling